

2 Elastin and collagen are proteins found in connective tissue.

(a) Describe the structure of collagen.

(3)

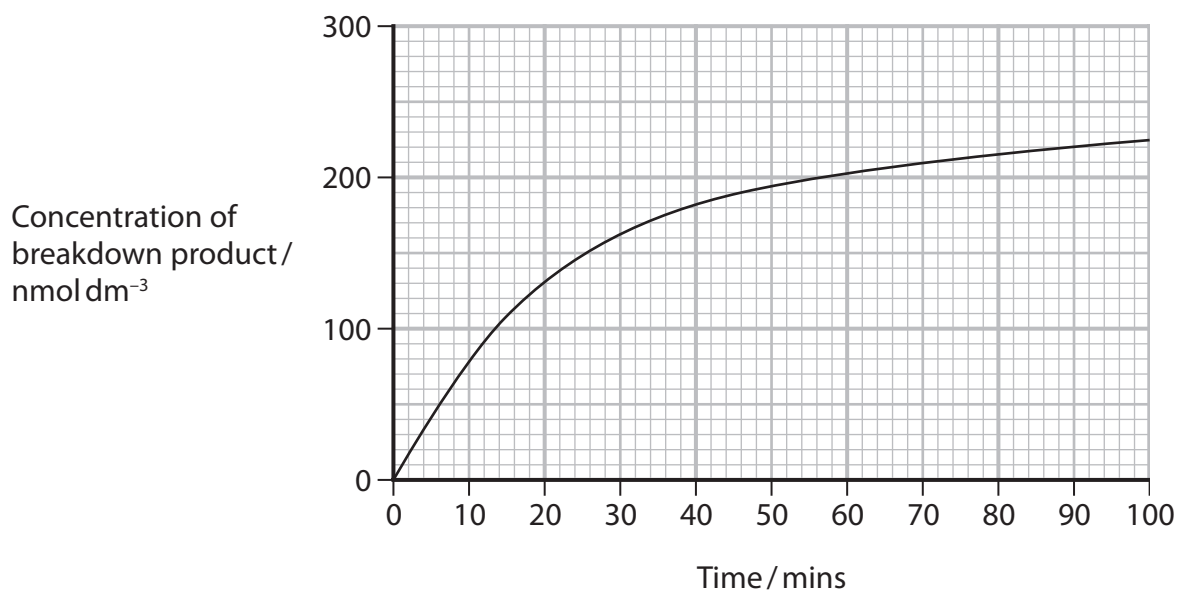
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(b) Proteases are enzymes that break down proteins.

The graph shows the effect of a protease solution on a sample of elastin.



- (i) Calculate the rate of reaction at 30 minutes.

Draw a tangent to help with your calculation.

(2)

Answer $\text{nmol dm}^{-3} \text{ min}^{-1}$

- (ii) Explain how a protease breaks down elastin.

(3)

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(Total for Question 2 = 8 marks)

